

A Cold Tracer in a Hot Bath: In and Out of Equilibrium

Amer Al-Hiyasat

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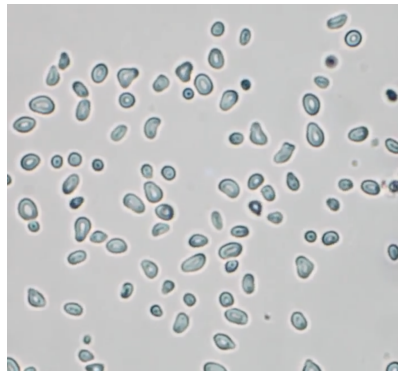
10 July, 2025



AA, S Ro, J Tailleur,
arXiv:2501.04930
arXiv:2503.01998

Equilibrium is contagious:

Tracer in **equilibrium** fluid has **equilibrium** dynamics.



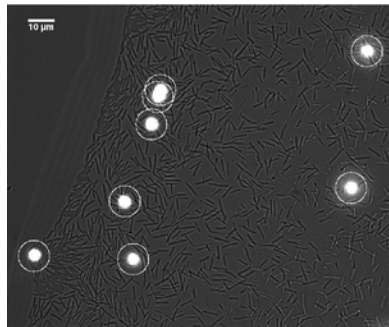
Pollen grains in water

(Koshu Endo)

Equilibrium is contagious:

Tracer in **equilibrium** fluid has **equilibrium** dynamics.

Is nonequilibrium contagious?



Colloids in a bath of *Bacillus subtilis*
(Valeriani et al.)

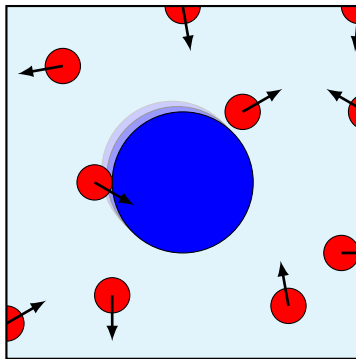
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Is nonequilibrium contagious? Yes:

Tracer in **active** bath behaves like **active** particle.

(Wu & Libchaber; Granek et al.; Solon & Horowitz; others)



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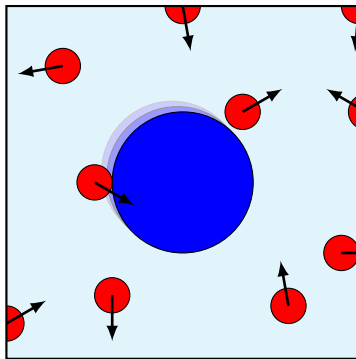
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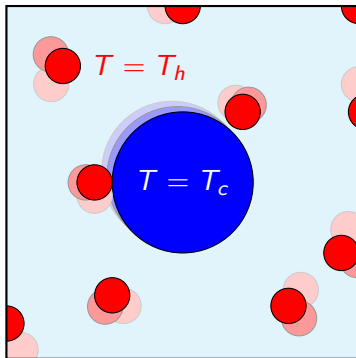
Tracer in **active** bath behaves like **active** particle.

(Wu & Libchaber; Granek et al.; Solon & Horowitz; others)

How generic is this?

Intermediate case: **cold tracer in hot bath.**

(Ilker et al, Wang et al., Jardat et al., Goswami et al., Schwarcz et al.)



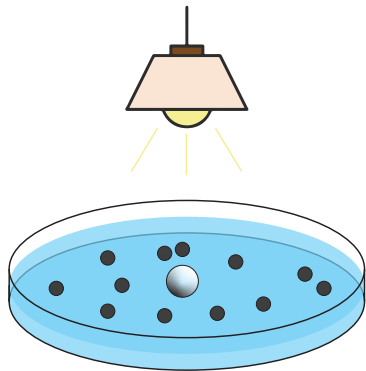
A cold tracer in a hot bath

Bath particles subject to **enhanced fluctuations**

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Bath particles subject to **enhanced fluctuations**

1. External irradiation



A cold tracer in a hot bath

Bath particles subject to **enhanced fluctuations**

1. External irradiation
2. Active enzyme solutions

NANO LETTERS

Letter

pubs.acs.org/NanoLett

Enhanced Diffusion of Passive Tracers in Active Enzyme Solutions

Xi Zhao,^{†,⊥} Krishna K. Dey,^{‡,⊥} Selva Jeganathan,[§] Peter J. Butler,[§] Ubaldo M. Córdova-Figueroa,^{*,||}
and Ayusman Sen^{*,†,⊕}

A cold tracer in a hot bath

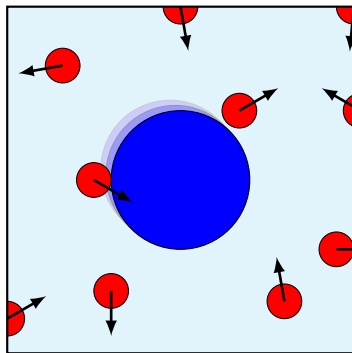
Bath particles subject to **enhanced fluctuations**

1. External irradiation
2. Active enzyme solutions
3. Self-propelled particles with scale separation

Colloid in bacterial bath:

Colloid: $T \sim 300\text{K}$

Bacteria: $T_{\text{eff}} \sim (5 \times 10^5)\text{K}$



A cold tracer in a hot bath

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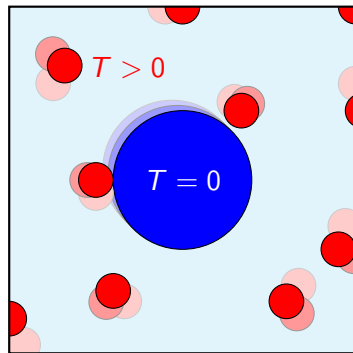
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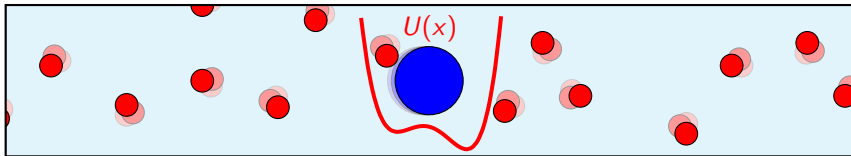
Bacteria: $T_{\text{eff}} \sim (5 \times 10^5)\text{K}$

Tracer: $\dot{\mathbf{x}} = -\mu \nabla_{\mathbf{x}} V(\mathbf{x}, \{\mathbf{x}_i\})$

Bath: $\dot{\mathbf{x}}_i = -\mu_i \nabla_{\mathbf{x}_i} V(\mathbf{x}, \{\mathbf{x}_i\}) + \sqrt{2\mu_i T} \boldsymbol{\eta}_i(t)$

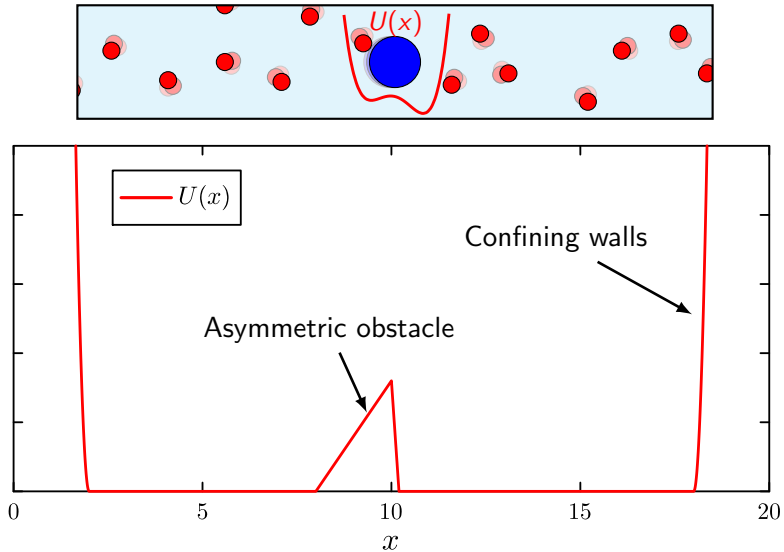


Tracer distribution in a potential

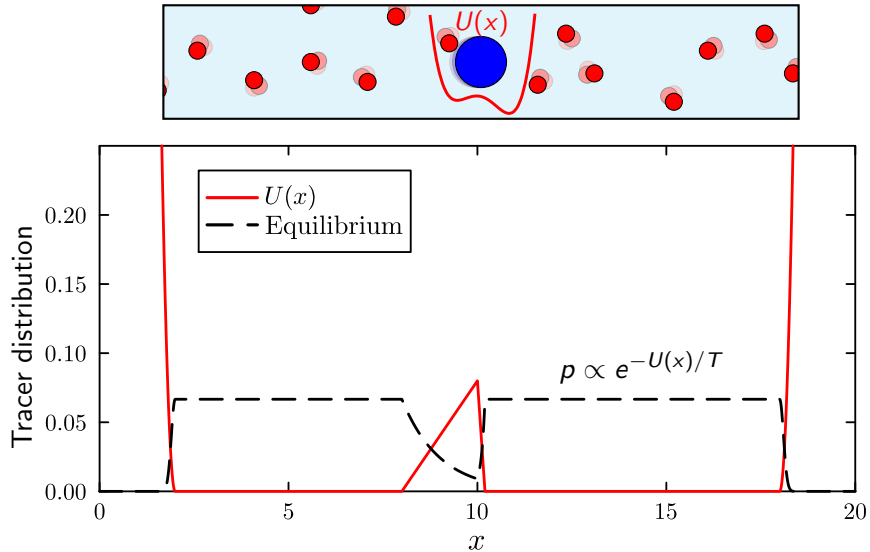


- ▶ Short-ranged repulsive interactions
- ▶ External potential on cold tracer $U(x)$

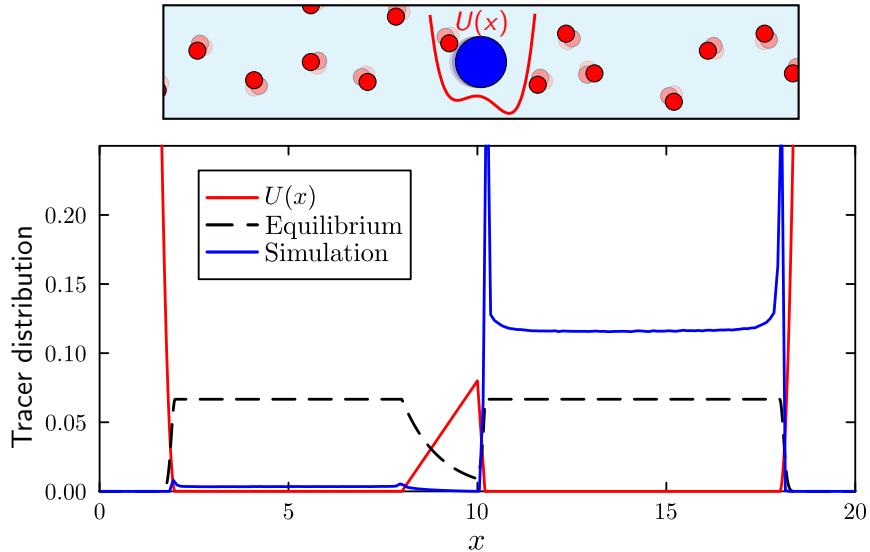
Tracer distribution in a potential



Tracer distribution in a potential

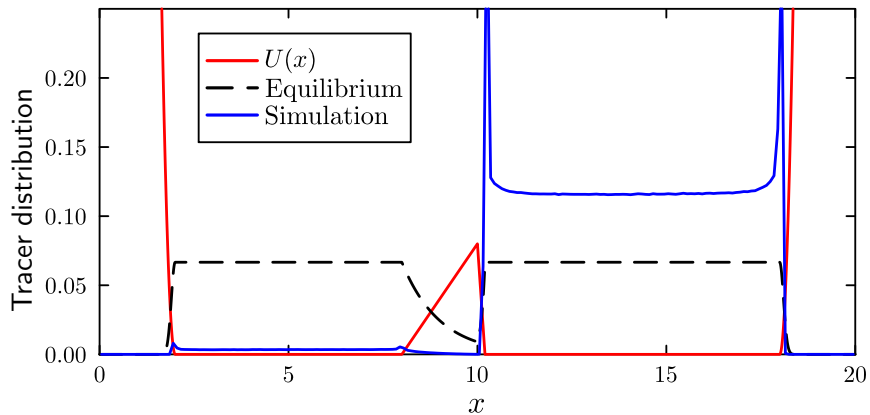


Tracer distribution in a potential



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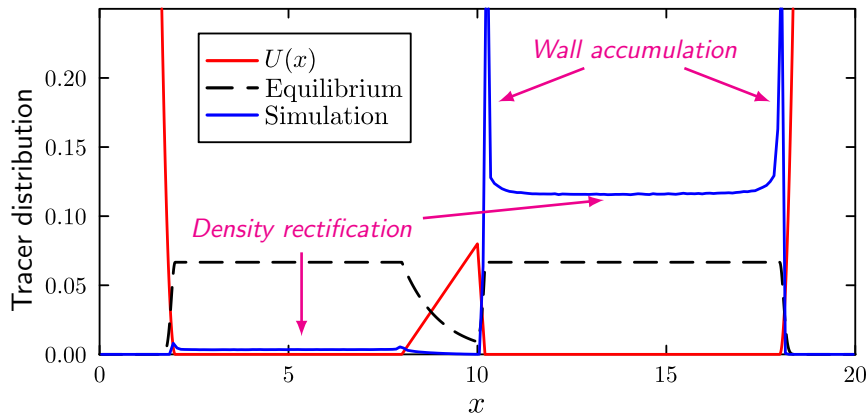
Conclusion 1: cold tracer in a hot bath has non-Boltzmann statistics



Tracer distribution in a potential

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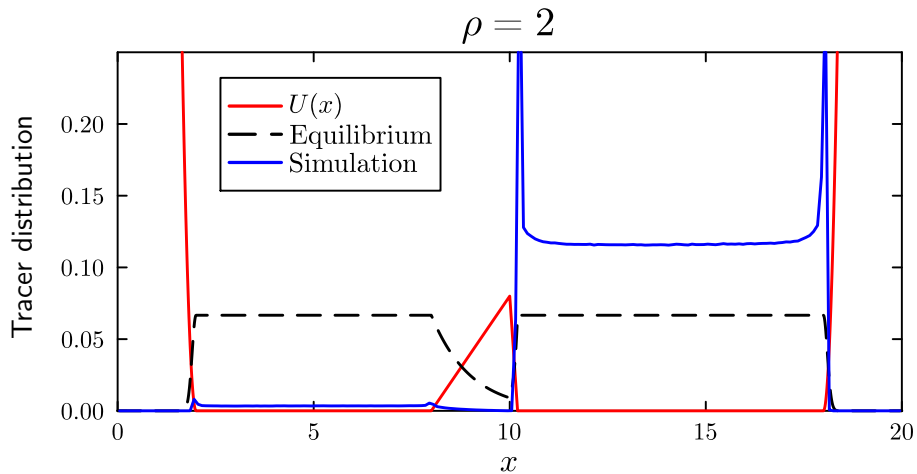
$\Rightarrow$ *Similar to active particle*



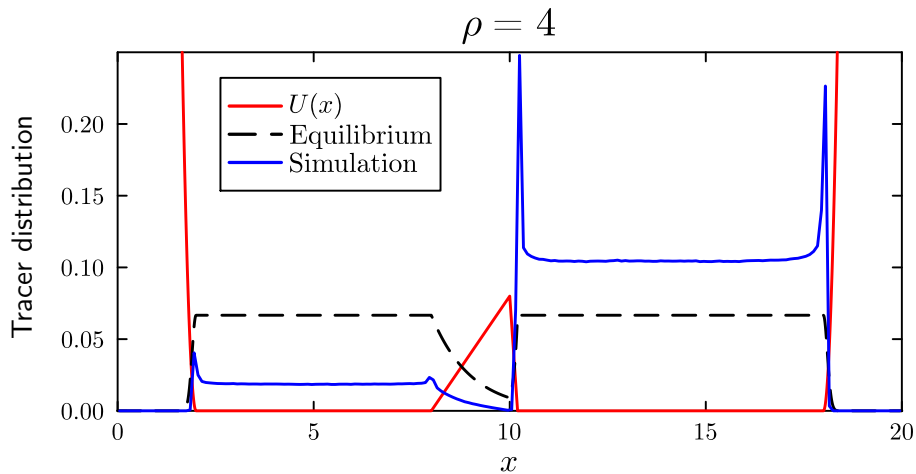
What if we increase the bath density?

$\rho \equiv$ **number of bath particles per unit tracer radius**

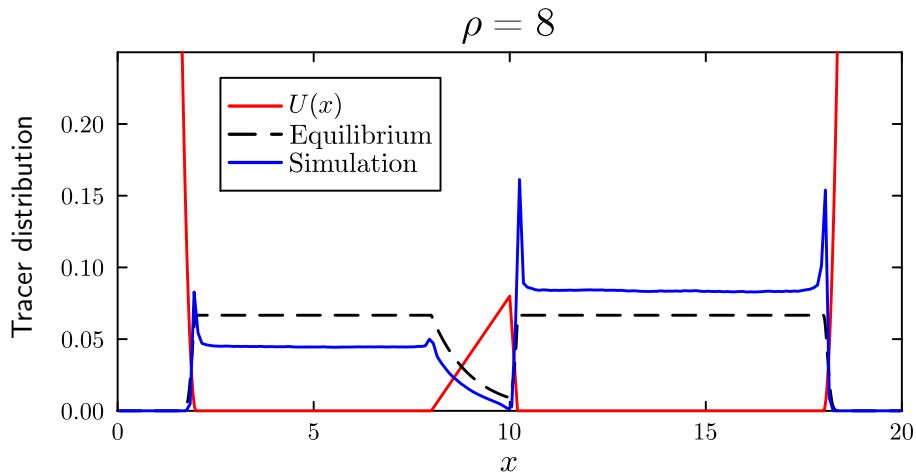
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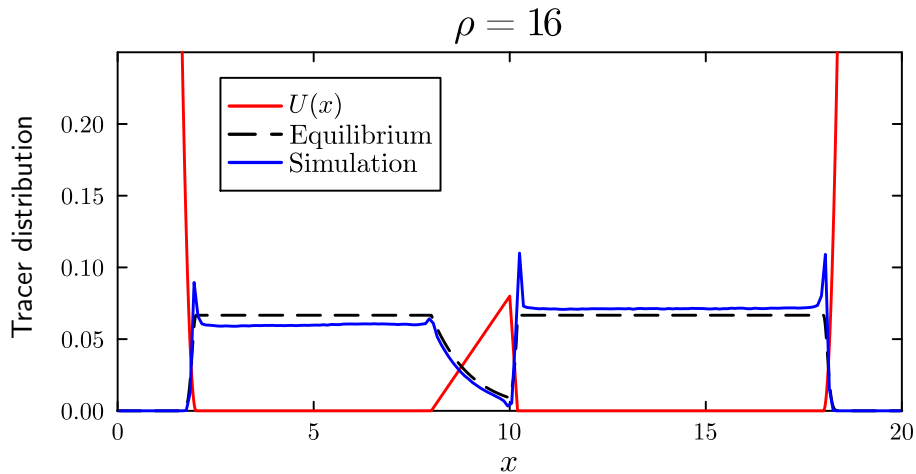
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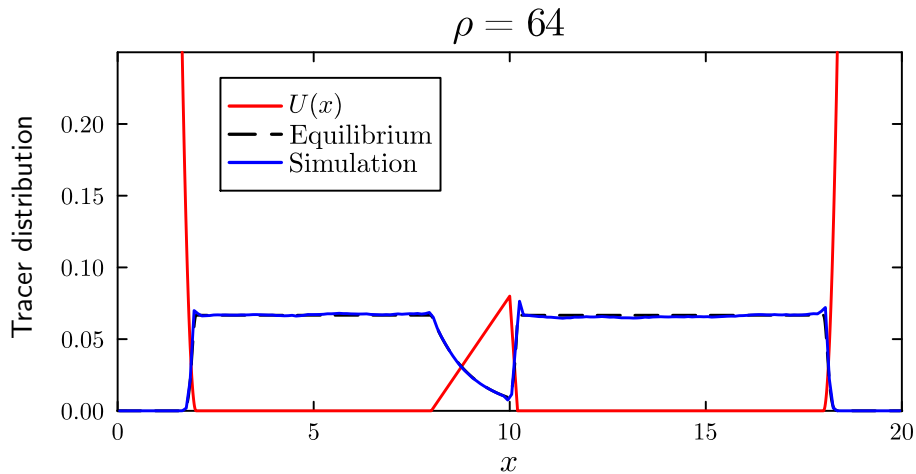
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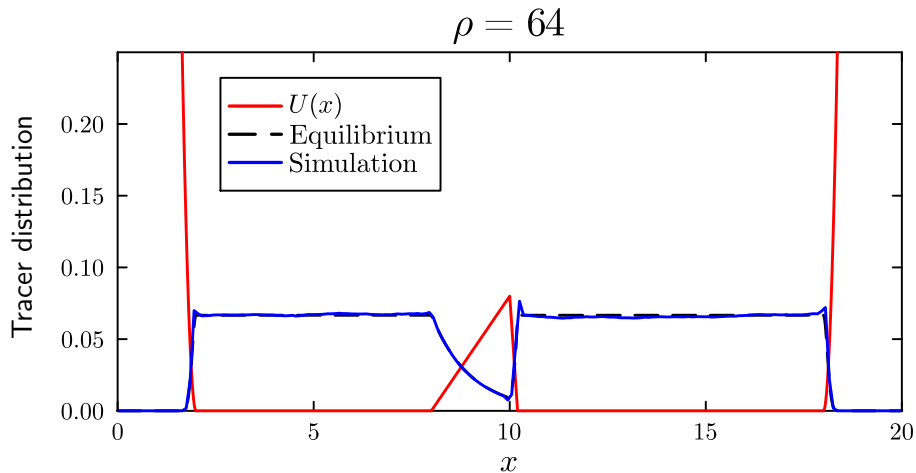
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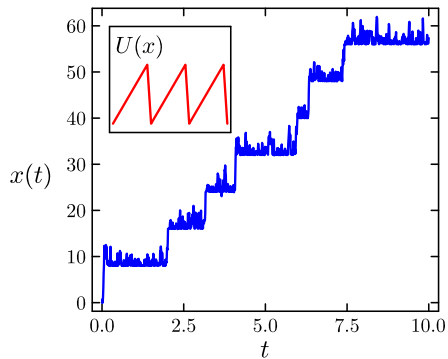
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Conclusion 2: nonequilibrium effects vanish in the $\rho \rightarrow \infty$ limit.

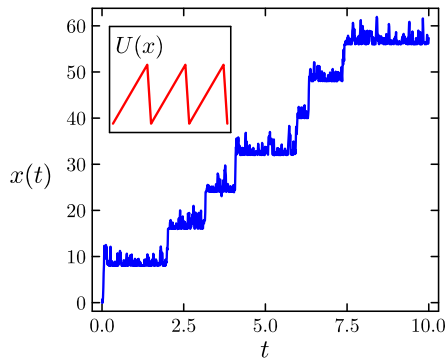
Ratchet currents in asymmetric, periodic potentials

Nonequilibrium dynamics at finite ρ

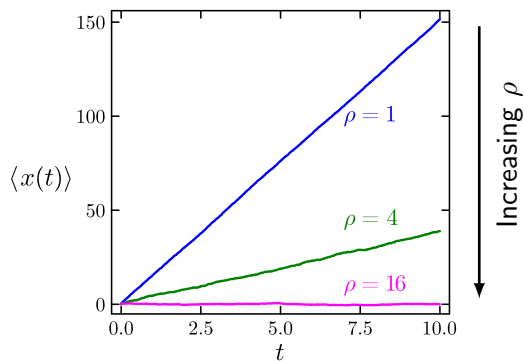


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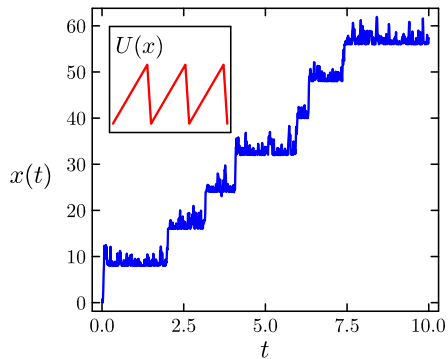


Equilibrium dynamics as $\rho \rightarrow \infty$

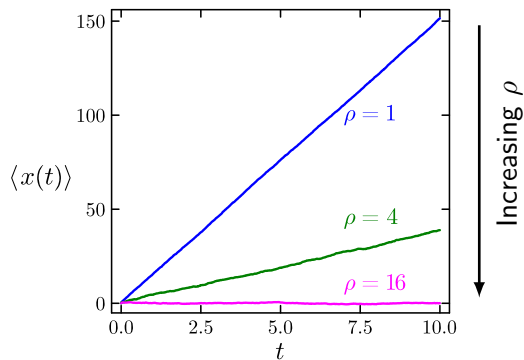


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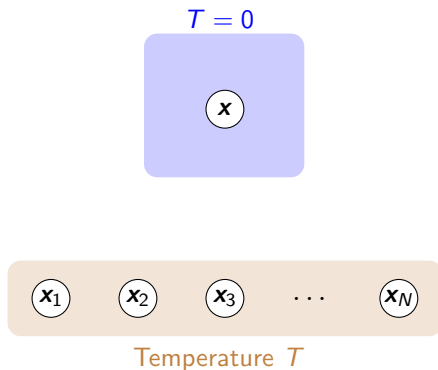
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Q: How do we explain this transition?

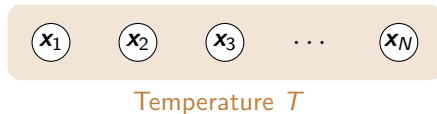
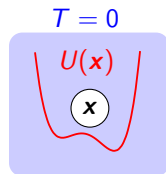
Fully-connected model

- ▶ $\mathbf{x}$: tracer at $T = 0$
- ▶ $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$: bath at temperature T



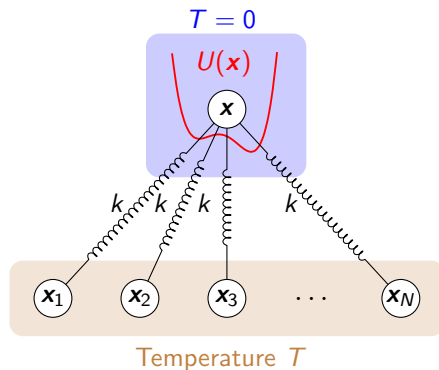
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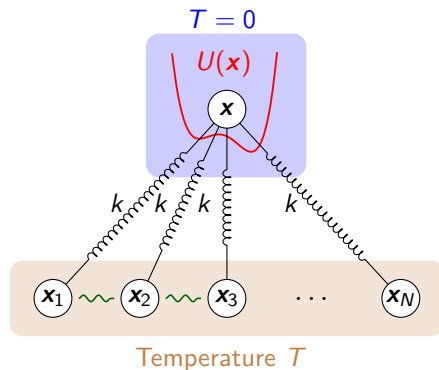
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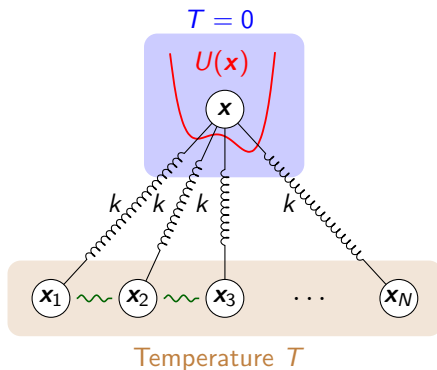
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For $N \gg 1$, leading order tracer dynamics:

$$\dot{\mathbf{x}} = -\frac{\nabla U(\mathbf{x})}{N} + \sqrt{\frac{2T}{N}}\eta(t),$$

$\Rightarrow$ **Equilibrium dynamics at temperature T with mobility $1/N$**



Finite N perturbation theory

- Eliminate bath degrees of freedom to obtain:

$$\dot{\mathbf{x}} = \mathbf{p}$$

$$\dot{\mathbf{p}} = -\nabla U - (\mathbf{p} \cdot \nabla) \nabla U - (N+1)\mathbf{p} + \sqrt{2TN}\boldsymbol{\eta}(t).$$

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- ▶ Large- N perturbation theory for stationary measure:

$$\partial_t \Psi(\mathbf{x}, \mathbf{p}) = \mathcal{L}\Psi(\mathbf{x}, \mathbf{p}) = 0, \quad \Psi(\mathbf{x}, \mathbf{p}) \equiv \frac{1}{Z} e^{-\beta \mathcal{H}_{\text{eff}}(\mathbf{x}, \mathbf{p})}.$$

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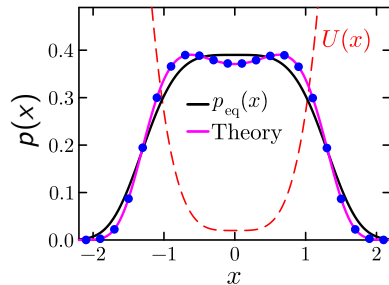
$$\mathcal{H}_{\text{eff}}(\mathbf{x}, \mathbf{p}) = U + \frac{p^2}{2} + \frac{1}{2N} \left[(\nabla U)^2 + 2U - 3T \nabla^2 U + (\mathbf{p} \cdot \nabla)^2 U + p^2 \right] + \mathcal{O}(N^{-2})$$

AA, S Ro, J Tailleur, arXiv:2501.04930 & arXiv:2503.01998

Departure from equilibrium at finite N

1. Deviation from Boltzmann at $\mathcal{O}(N^{-1})$

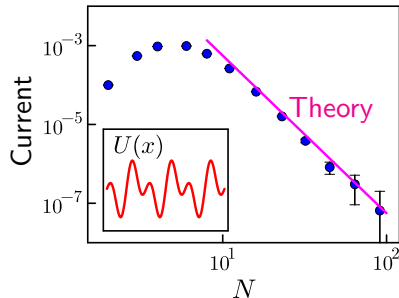
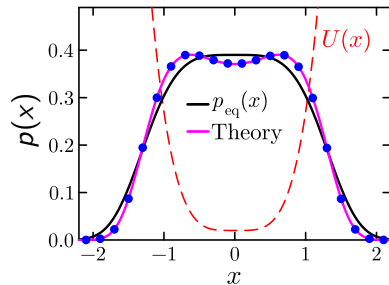
$$p(\mathbf{x}) \propto \exp \left\{ -\frac{U}{T} - \frac{1}{N} \left[\frac{(\nabla U)^2}{2T} - \nabla^2 U + \frac{U}{T} \right] + \mathcal{O}\left(\frac{1}{N^2}\right) \right\}$$



Departure from equilibrium at finite N

1. Deviation from Boltzmann at $\mathcal{O}(N^{-1})$
2. Ratchet currents $\propto N^{-4}$

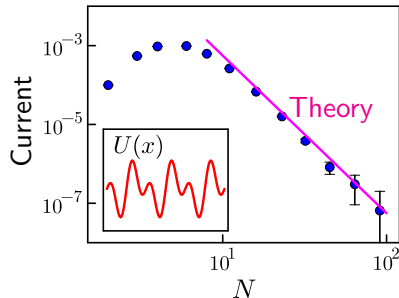
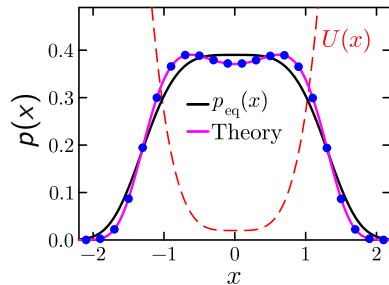
$$\langle \dot{x} \rangle = \frac{L \int_0^L U'(z)^2 U^{(3)}(z) dz}{2N^4 \int_0^L e^{-\frac{U(y)}{T}} dy \int_0^L e^{\frac{U(z)}{T}} dz} + \mathcal{O}(N^{-5})$$



Departure from equilibrium at finite N

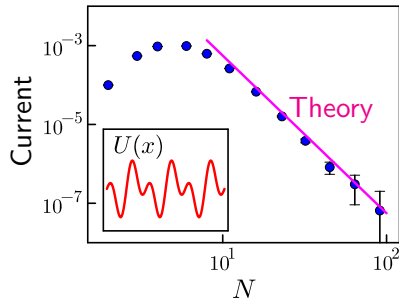
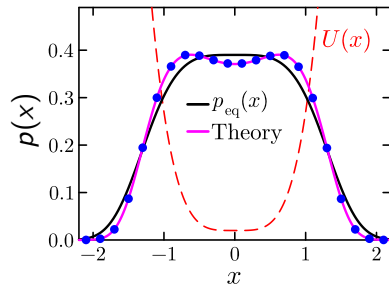
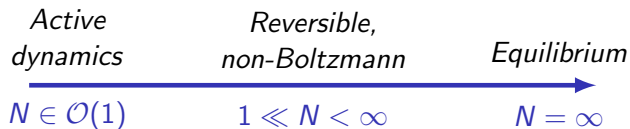
1. Deviation from Boltzmann at $\mathcal{O}(N^{-1})$
2. Ratchet currents $\propto N^{-4}$
3. Entropy production rate $\propto N^{-3}$

$$\sigma = \frac{T}{2N^3} \left\langle (\nabla^3 U)^2 \right\rangle_{\text{eq}} + \mathcal{O}(N^{-4})$$

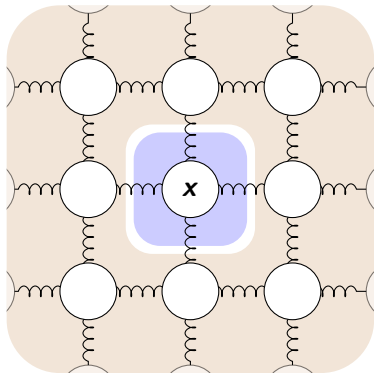
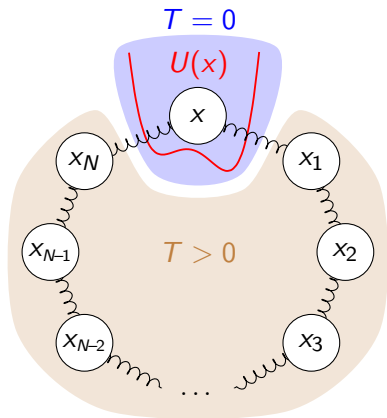


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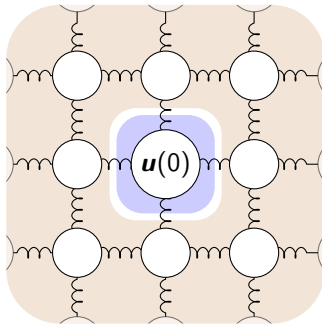
A cold tracer in a hot “gel”



Coarse-grained field theory

- Overdamped elastodynamics

$$\partial_t \mathbf{u}(\mathbf{r}, t) = \nabla^2 \mathbf{u} + \kappa \nabla (\nabla \cdot \mathbf{u}) + \mathbf{\Lambda}(\mathbf{r}, t) .$$

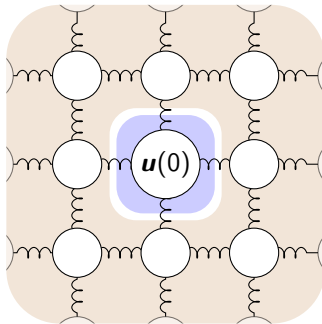


Coarse-grained field theory

- Overdamped elastodynamics with **cold inclusion**:

$$\partial_t \mathbf{u}(\mathbf{r}, t) = \nabla^2 \mathbf{u} + \kappa \nabla (\nabla \cdot \mathbf{u}) + \mathbf{\Lambda}(\mathbf{r}, t) .$$

$$\langle \Lambda_i(\mathbf{r}, t) \Lambda_j(\mathbf{r}', t') \rangle = 2T \delta_{ij} \delta(t - t') \delta(\mathbf{r} - \mathbf{r}') \left[1 - L^{-d} \delta(\mathbf{r}) \right]$$



Coarse-grained field theory

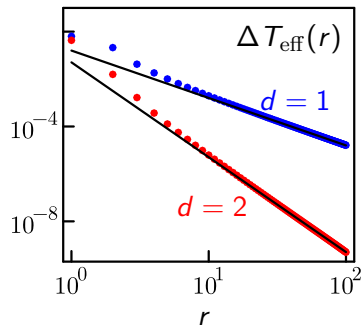
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- Result: **Long-ranged suppression of fluctuations**

$$\Delta T_{\text{eff}}(\mathbf{r}) \sim r^{-2d}$$



Coarse-grained field theory

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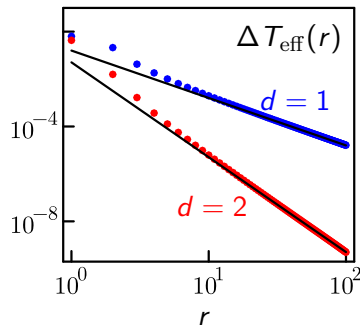
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- ▶ Result: **Long-ranged suppression of fluctuations**

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Conclusion: *cold tracer cools the entire lattice*



Conclusions

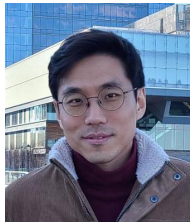
- ▶ Cold tracer in hot fluid: active at small bath density; equilibrium at large density.
- ▶ Cold tracer in hot lattice: nonequilibrium dynamics; long-ranged cooling of lattice.

Conclusions

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Prof. Julien Tailleur
(MIT)



Prof. Sunghan Ro
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Preprints:

- ▶ AA, S Ro, J Tailleur, arXiv:2501.04930
- ▶ AA, S Ro, J Tailleur, arXiv:2503.01998

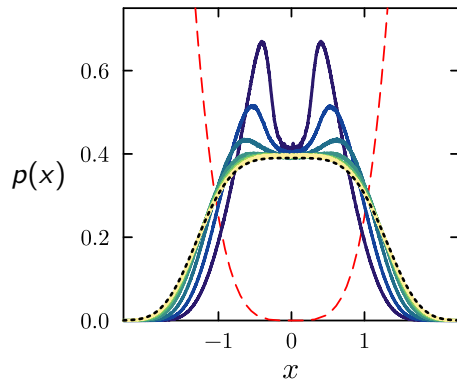
Funding:

- ▶ MIT presidential fellowship
- ▶ Mathworks Science Fellowship
- ▶ MISTI and ICAM travel grants

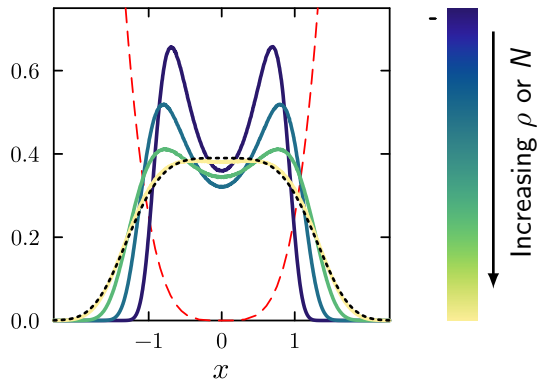
Comparing short-ranged and fully-connected models

Apply quartic potential to tracer $U(x) \propto x^4$

Short-ranged model



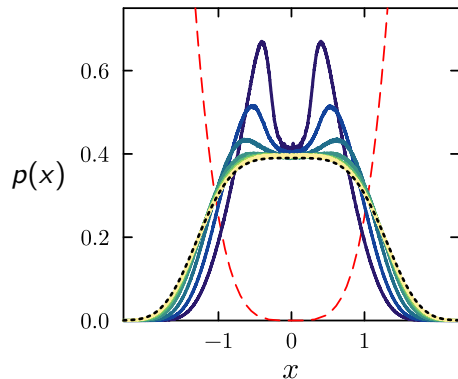
Fully-connected model



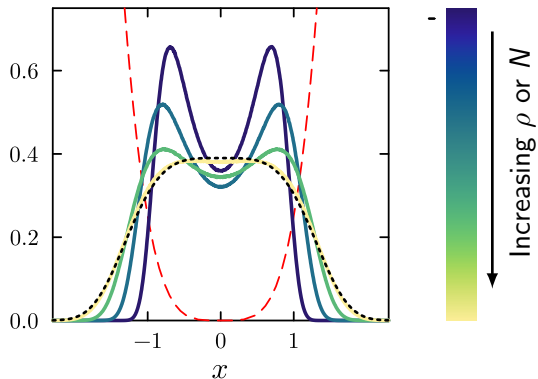
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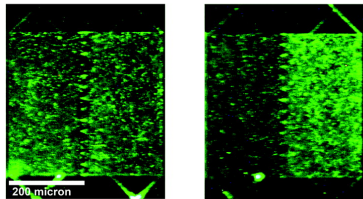
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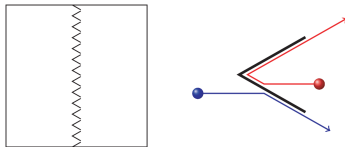
Large N in fully-connected model $\Leftrightarrow$ Large ρ in short-ranged model

Similarity to self-propelled particles

Rectification of *E. coli* by asymmetric funnels



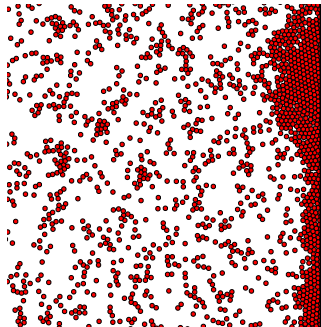
(Galajda et al.)



(Tailleur & Cates)

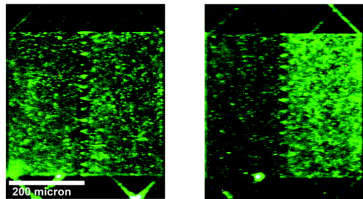
Wall accumulation of self-propelled particles

(Elgeti & Gompper)

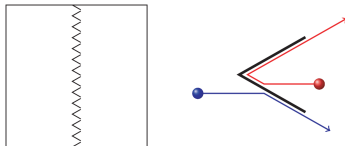


Similarity to self-propelled particles

Rectification of *E. coli* by asymmetric funnels



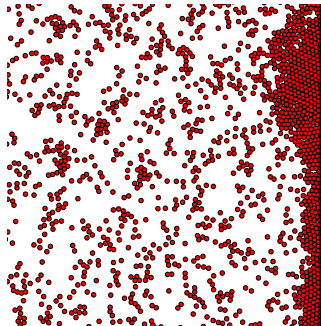
(Galajda et al.)



(Tailleur & Cates)

Wall accumulation of self-propelled particles

(Elgeti & Gompper)



Cold particle in hot bath $\Leftrightarrow$ Active particle

Generalized Langevin equation

- Can show that the tracer obeys:

$$\int_{-\infty}^t ds K(t-s) \dot{x}(s) = -U'(x) + \zeta(t)$$

where

$$K(s) = 2\mu^{-1}\delta(s) + Nke^{-k|s|}, \quad \langle \zeta(t)\zeta(t+s) \rangle = NkTe^{-k|s|}$$

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- ▶ FDT is satisfied iff $\langle \zeta(t)\zeta(t+s) \rangle = TK(s)$
- ▶ Rescale time by $N \Rightarrow$ get FDT as $N \rightarrow \infty$